Explain the concept of Cloud Computing with suitable examples to show its practical significance.
Describe the major characteristics of Cloud Computing and their importance in modern IT infrastructure.
Discuss the evolution of Cloud Computing from traditional computing models to the present cloud-based environment.
Explain the role of utility computing, grid computing, and virtualization in the evolution of Cloud Computing.
Describe the concept of Parallel Computing in cloud environments along with its advantages.
Explain Distributed Computing in the context of cloud systems and illustrate its working with a suitable example.
Differentiate between Parallel and Distributed Cloud systems with appropriate explanation.
Discuss the fundamental principles of Cloud Computing such as on-demand self-service, resource pooling, scalability, and measured service.
Explain the concept of REST architecture and describe its key constraints in web-based applications.
Describe the concept of System of Systems and discuss its characteristics in large-scale computing environments.
Explain the structure and functioning of Web Services in distributed systems.
Describe the basic concept of Virtualization and illustrate its significance in modern computing infrastructure.
Explain different types of Virtualization such as full virtualization, para-virtualization, and container-based virtualization.
Describe various levels of Virtualization including hardware level, operating system level, and application level.
Explain the tools and mechanisms used in Virtualization with suitable examples.
Discuss the role of Virtualization in disaster recovery and system backup strategies.
Elaborate the design of Layered Cloud Architecture by describing its different layers, their functions, and interaction among infrastructure, plat
Discuss the NIST Cloud Computing Reference Architecture by explaining its key actors, core components, and service interaction model in det
Describe Public, Private, and Hybrid Cloud deployment models by presenting their structural characteristics, operational environment, and suit
Explain the comparative features of Public, Private, and Hybrid Cloud models by interpreting aspects such as control, scalability, cost efficienc
Illustrate the differences between IaaS, PaaS, and SaaS with suitable examples.
Demonstrate how Cloud Storage works and list its advantages over traditional storage.
Apply the concept of Storage-as-a-Service to explain how businesses can use cloud storage for their data needs.
Implement the features of Amazon S3 and explain how it is used as a cloud storage provider.

10 Marks

Elaborate the concept of Cloud Computing by describing its architecture, essential components, and service delivery environment in detail.
Discuss the historical development of Cloud Computing by outlining the transition from mainframe systems to virtualized and service-oriented
Explain the relationship between virtualization technology and the growth of Cloud Computing with suitable illustrations.
Describe the operational model of Parallel Cloud systems and interpret their significance in handling large-scale computational problems.
Explain the structure and functioning of Distributed Cloud environments and summarize their advantages over centralized computing systems.
Discuss the core principles governing Cloud Computing and interpret their role in ensuring flexibility, cost efficiency, and optimized resource u
Elaborate REST-based web services by describing their architectural style, resource representation, HTTP methods, and stateless communicat
Discuss the concept of System of Systems by outlining its structural components, coordination mechanisms, and integration challenges in dis
Explain the complete lifecycle of a Web Service including service creation, description, publishing, discovery, and communication process.
Describe the architecture of Virtualization by explaining the interaction between host machine, hypervisor, guest operating systems, and virtua
Discuss various types and levels of Virtualization by presenting their working principles, advantages, and limitations in enterprise environments
Elaborate the mechanisms supporting Virtualization such as live migration, load balancing, snapshot management, and disaster recovery strat
Explain the structural framework of Layered Cloud Architecture by illustrating the role of service layer, resource layer, and management layer i
Describe the operational workflow defined in the NIST Cloud Computing Reference Architecture by presenting the interaction among cloud pr
Discuss the integration model of Hybrid Cloud by explaining workload distribution, data management strategy, and coordination between publi
Illustrate the architectural design challenges in cloud computing. Demonstrate how IaaS, PaaS, and SaaS address these challenges with exam
Implement a detailed explanation of Cloud Storage and Storage-as-a-Service. Apply this understanding to compare major cloud storage provi
Demonstrate the complete service model of cloud computing covering IaaS, PaaS, and SaaS. Apply suitable examples to highlight the advanta